Fangjie Gao

+1 734-773-7054 | fgao76@gatech.edu | Atlanta, GA

EDUCATION

Georgia Institute of Technology — M.S. Electrical and Computer Engineering Fall 2026
Courses: Convex Optimization, Data Mining & Statistical Learning, Mathematical Foundations of Machine Learning, Conversational AI

University of Michigan — B.S. Industrial and Operations Engineering, Minor in EE May 2026 · GPA 3.84/4.00
Courses: Statistical Quality Control, Operations Analysis & Methods, Discrete-Event Simulation, Derivative Instruments, Intro to Data Analytics, Optimization and Computational Methods, Data Structures and Algorithms, Computer Organization

Shanghai Jiao Tong University — B.S. Electrical and Computer Engineering August 2026 · GPA 3.37/4.00
Courses: Analog Electronic Circuits, Signals and Systems, Electromagnetics, Linear Algebra, Discrete Mathematics, Probabilistic Methods

TECHNICAL SKILLS

ML: prompt engineering, RAG, LangChain, predictive labeling, cross-validation, backtesting, workflow automation, evaluation, hyperparameter tuning, anomaly detection
Programming: Python, pandas, SQL, C, C++, MATLAB, Shell
Infrastructure: Docker, Microsoft, GitHub Actions, Git, SSH, Nginx, n8n, HuggingFace Spaces, pybind11, Tmux, Vim, Make, Power BI
Languages: Fluent English (TOEFL 111), Native Chinese

SUMMARY

Circuit design background spanning analog, RF, and digital domains. Switched focus to data analysis and agent. Experienced in data-driven process improvement, low-latency optimization, and AI agent workflows.

WORK EXPERIENCE

Shanghai Banzhi Quant Technology Co., Ltd — Quantitative Development Intern May – Jul 2026

- Built Python-C++ trading interface wrappers for the CTP (China Futures Trading Platform) using pybind11; profiled and optimized latency across the Python-C++ request-callback chain using flame graph analysis, with CPU affinity pinning.
- Diagnosed and removed a hot-path double-copy marshaling layer, replaced a generic raw dict-to-dynamic-class reconstruction (~55 setattr calls per callback) with direct typed struct-to-dict conversion.
- Built end-to-end latency instrumentation (5 trace checkpoints, 4 stage metrics) spanning the Python, C++, exchange dynamic library boundary, plus a benchmark harness and trimmed-mean visualization tooling.
- Performed live A/B testing across 3 independent trading sessions; callback-processing latency down ~61% single-process, ~86% multi-process, consistent in all 6 of 6 comparisons.
- Conducted system testing and validation in a local simulated trading exchange environment.

Interventional Radiology Process Optimization — IOE 481 Practicum, Michigan Medicine Jan – Apr 2026

- Ran weekly stakeholder meetings, writing agendas and sending recaps/action plans to keep the team aligned with client priorities.
- Conducted 30-minute interviews with 3 stakeholder groups (IR surgeon, procedure nurses, technologist supervisors) and 4 on-site process observations (7:00AM–12:00PM) to map the first-case workflow and cross-validate root causes.
- Analyzed a full-year (2025) delay dataset, quantifying $1.1M+ in annual loss ($57/min IR downtime) from 700+ delays coded "Unknown" (55% of entries); ranked the Top 10 delay drivers by frequency × severity via Pareto analysis.
- Prioritized data quality, eliminated "unknown" delay-reason entries, and categorized delay comments to isolate the patient class with the most uncertainty.
- Standardized SOAP to reduce consent-related delays and synthesized findings into current-state/future-state process maps and a formal report presented to IR operations leadership.

Inceptio Technology — Data Analysis Intern Aug 2023

- Analyzed 6 months of transportation data from 200+ trucks (~50K trips) using SQL to extract relationships between fuel consumption and traffic congestion, wind resistance, payload, and route topology.
- Built Python visualizations (correlation heatmaps, partial dependence plots) to identify 5 statistically significant factors; congestion and wind resistance accounted for 35% of fuel consumption variance.
- Ranked factors by impact and feasibility, identifying route optimization as the highest-ROI lever with potential to reduce fuel consumption by 8–12%.

PROJECT EXPERIENCE

Credit Card Fraud Detection (Kaggle in progress) Aug 2026 - Now

- Built a feature engineering pipeline that transforms raw transaction records into 62 features, covering temporal, geospatial, behavioral, and categorical signals
- Engineered per-card behavioral features, including rolling transaction counts and spend over 1h–7d windows, spending baselines and z-scores computed strictly from prior transactions, and Haversine-based "impossible travel" detection between consecutive transactions
- Encoded high-cardinality fields (merchant, job, city) with out-of-fold, Bayesian-smoothed target encoding, and added first-time merchant/category flags to capture deviations from a cardholder's routine

Self-Hosted AI Agent Music Management System May – Aug 2026

Stack: Python, FastAPI, WebSocket, React, LangChain, n8n, Docker, Nginx, systemd

- Architected a conversational AI agent using LangChain tool-calling, enabling natural language intent recognition to auto-route requests to backend tools, achieving 95%+ tool-invocation accuracy across 4 core operations.
- Built a real-time backend with FastAPI + WebSocket, implementing a streaming status-update pipeline that decomposes 15-30s multi-step tasks into visible progress states.
- Deployed a self-hosted LLM gateway to unify multi-provider model access, centralizing API key management and enabling flexible model switching and cost comparison.
- Automated an end-to-end audio acquisition pipeline using n8n for batch download tasks.
- Designed a fault-tolerant parsing layer to work around third-party API limitations — replaced schema-constrained generation with example-guided prompting and manual JSON parsing, improving batch task success rate from 0% to 95%+, and resolved output-truncation failures via batched processing.
- Implemented defensive security: application-level path whitelist validation to prevent directory traversal, a dedicated limited-privilege account, and systemd sandboxing restricting filesystem access to 2 whitelisted directories.

AI-Supported Quantitative Research Platform Jan – Aug 2026

- Designed and built a full-stack AI quantitative research platform integrating multi-source data into an end-to-end pipeline from ingestion through technical analysis to LLM-generated reports.
- Implemented 10+ classical technical-analysis systems in Python.
- Engineered a prompt pipeline structuring factors, macro data, news, and retail sentiment for an LLM (Groq Llama-3.3-70B) to auto-generate research reports.
- Containerized and deployed via Docker to HuggingFace Spaces.
- Built a fully automated data pipeline, including ingesting, LLM analysis, HTML rendering, email delivery, and containerized with Docker and secured via GitHub Secrets for credential management.